

Laplace-Domain Analysis of Fractional RLC Circuits: A Reproducible Study of Transient and Frequency Responses

Abstract—This paper studies a series RLC circuit using the ordinary Laplace transform and Caputo fractional derivatives. A dimensional time scale preserves consistent element units and recovers the classical circuit when the derivative order equals one. A reproducible numerical scheme compares capacitor voltage, circuit current, frequency response, and settling behavior for three orders. The results show that reduced overshoot can coexist with slower final convergence, so settling depends on the chosen tolerance. Classical analytical solutions and numerical Laplace inversion check the computed responses. The contribution is an undergraduate comparison that connects transform theory, circuit modeling, and verified simulation without experimental claims.

Index Terms—RLC circuit, Laplace transform, Caputo derivative, fractional order, MATLAB, signals and systems.

I. INTRODUCTION

The series resistor–inductor–capacitor (RLC) circuit is a useful bridge between circuit theory and signals and systems. Kirchhoff’s laws lead to a second order differential equation; its Laplace transform produces a transfer function that predicts transient response, resonance, and filtering. An underdamped circuit also demonstrates why stability alone does not guarantee a small overshoot or a short settling time. These features make the RLC circuit an appropriate starting point for investigating models with memory.

Fractional derivatives generalize differentiation to a noninteger order. Their integral definitions involve the previous trajectory, allowing a compact description of distributed relaxation and constant phase behavior [2], [3]. Such models are used for nonideal electrical elements and circuit approximations. An ideal laboratory resistor, inductor, and capacitor still obey the usual integer order laws. A fractional circuit therefore requires an explicit constitutive model; changing a derivative order in a familiar equation without specifying units and initial conditions is insufficient.

Moazzam *et al.* discuss electrical applications of a “Fractional-Laplace” transformation [1]. Their transform has kernel $\exp[-v^{1/\alpha}t]$, which becomes the ordinary Laplace kernel after setting $s = v^{1/\alpha}$. This change of transform variable is useful algebraically, but does not itself introduce physical memory into an ideal RLC circuit. The present paper uses the ordinary unilateral Laplace transform of Caputo derivatives and independently derives the circuit model. This distinction keeps the physical assumptions separate from the transform notation.

The contribution is a reproducible course level comparison, rather than a claim of a new fractional calculus theorem or

measured device performance. It provides a dimensionally consistent model, verifies recovery of the classical response, checks a full history discretization against independent Laplace inversion, and compares two settling tolerances. The central question is whether a smaller fractional order improves all response measures at once. Voltage, current, and frequency curves answer this question for one fixed, clearly specified circuit.

II. CLASSICAL CIRCUIT AND REFERENCE SOLUTION

A. Circuit equations and normalization

The source drives a series circuit, with the voltage across the capacitor selected as the output (Fig. 1). Kirchhoff’s voltage law and the capacitor law give

$$v_{in}(t) = L \frac{di}{dt} + Ri(t) + v_C(t), \quad (1)$$

$$i(t) = C \frac{dv_C}{dt}. \quad (2)$$

Eliminating the current produces

$$LC \frac{d^2 v_C}{dt^2} + RC \frac{dv_C}{dt} + v_C = v_{in}. \quad (3)$$

All simulations start from $v_C(0) = i(0) = 0$ and use a $V_0 = 1$ V step. The chosen values are $L = 10$ mH, $C = 100$ μ F, and $R = 10$ Ω . They describe an accessible classical reference, rather than parameters fitted to measured data.

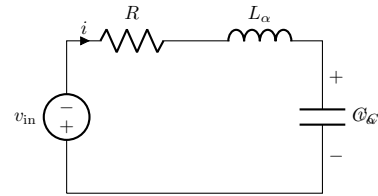


Fig. 1. Series topology. At $\alpha = 1$, $L_\alpha = L$ and $C_\alpha = C$. For $\alpha < 1$, the symbols denote the generalized elements defined in (10).

Define the time and impedance scales

$$\tau = \sqrt{LC} = 1 \text{ ms}, \quad Z_0 = \sqrt{L/C} = 10 \text{ } \Omega, \quad (4)$$

and the dimensionless variables

$$u = t/\tau, \quad x = v_C/V_0, \quad y = Z_0 i/V_0, \quad \rho = R/Z_0 = 1. \quad (5)$$

For the classical circuit, $x' = y$ and $y' = 1 - x - \rho y$. Thus the damping ratio is $\zeta = \rho/2 = 0.5$ and the undamped natural frequency is $\omega_n = 1/\tau = 1000$ rad/s. The normalized time u numerically equals time expressed in milliseconds for this particular choice of τ .

B. Exact integer order benchmark

Taking the unilateral Laplace transform of (3) under zero initial conditions gives

$$H_1(s) = \frac{V_C(s)}{V_{in}(s)} = \frac{1}{LCs^2 + RCs + 1}. \quad (6)$$

With $\omega_d = \sqrt{3}/2$ in normalized time, the exact step responses are

$$x_1(u) = 1 - e^{-u/2} \left[\cos(\omega_d u) + \frac{\sin(\omega_d u)}{\sqrt{3}} \right], \quad (7)$$

$$y_1(u) = \frac{2}{\sqrt{3}} e^{-u/2} \sin(\omega_d u). \quad (8)$$

The exact first voltage peak occurs at $u_p = 2\pi/\sqrt{3} = 3.6276$, with 16.3034% overshoot. These values provide a demanding numerical check because backward Euler integration introduces artificial damping. A solver that substantially suppresses this peak on a coarse grid cannot support a fair comparison with a fractional model.

III. FRACTIONAL CONSTITUTIVE MODEL

A. Caputo derivative and element units

For an absolutely continuous function and $0 < \alpha < 1$, the Caputo derivative is [2]

$${}^C D_t^\alpha f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{f'(\xi)}{(t-\xi)^\alpha} d\xi. \quad (9)$$

Its units are those of f multiplied by $s^{-\alpha}$. At $\alpha = 1$, the ordinary first derivative is used. The chosen generalized element laws are

$$\begin{aligned} v_L(t) &= L\tau^{\alpha-1} {}^C D_t^\alpha i(t), \\ i(t) &= C\tau^{\alpha-1} {}^C D_t^\alpha v_C(t). \end{aligned} \quad (10)$$

Consequently $L_\alpha = L\tau^{\alpha-1}$ has units $\text{H s}^{\alpha-1}$ and $C_\alpha = C\tau^{\alpha-1}$ has units $\text{F s}^{\alpha-1}$. These are generalized coefficients, not ordinary inductance and capacitance values when $\alpha < 1$. The factor $\tau^{\alpha-1}$ prevents addition of terms with incompatible dimensions and defines a fixed reference scale for comparing orders.

Both elements share the same order, giving a commensurate model. This assumption keeps the derivation and simulation suitable for an undergraduate study. It does not imply that a physical inductor and capacitor must have identical measured orders. A later identification study could fit separate orders and coefficients over a finite frequency band.

B. State equations and Laplace transfer function

Using Kirchhoff's law $v_{in} = v_L + Ri + v_C$, the scaling property ${}^C D_t^\alpha f(t) = \tau^{-\alpha} {}^C D_u^\alpha f(u)$, and (10) gives

$${}^C D_u^\alpha x = y, \quad {}^C D_u^\alpha y = r(u) - x - \rho y, \quad (11)$$

where $r = v_{in}/V_0$. In matrix form,

$${}^C D_u^\alpha \mathbf{z} = A\mathbf{z} + br, \quad A = \begin{bmatrix} 0 & 1 \\ -1 & -\rho \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad (12)$$

with $\mathbf{z} = [x \ y]^T$ and $\mathbf{z}(0) = 0$. For $0 < \alpha \leq 1$, the Caputo transform rule is

$$\mathcal{L}\{{}^C D_t^\alpha f\} = s^\alpha F(s) - s^{\alpha-1} f(0). \quad (13)$$

Applying it to both state equations yields

$$H_\alpha(s) = \frac{1}{(s\tau)^{2\alpha} + \rho(s\tau)^\alpha + 1}. \quad (14)$$

The principal branch of s^α is used. The current transfer function is

$$\frac{I(s)}{V_{in}(s)} = \frac{(s\tau)^\alpha}{Z_0} H_\alpha(s). \quad (15)$$

Setting $\alpha = 1$ exactly recovers $H_1(s)$. Equation (14) is derived from the state system; it does not require the generally invalid identity ${}^C D^\alpha {}^C D^\alpha x = {}^C D^{2\alpha} x$. Nonzero initial conditions would require retaining the separate initial terms in (13).

C. Stability and physical interpretation

The eigenvalues of A for $\rho = 1$ are

$$\lambda_{1,2} = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}, \quad |\arg \lambda_{1,2}| = \frac{2\pi}{3}. \quad (16)$$

The commensurate fractional stability condition is $|\arg \lambda_k| > \alpha\pi/2$ [5]. It holds for all three tested orders, $\alpha \in \{0.8, 0.9, 1.0\}$. The equilibrium under a constant unit input is $x_\infty = 1, y_\infty = 0$; stability therefore establishes the final voltage and current without assuming an exponential transient.

For positive frequencies, the generalized impedances are

$$Z_L = L\tau^{\alpha-1} (j\omega)^\alpha, \quad Z_C = \frac{1}{C\tau^{\alpha-1} (j\omega)^\alpha}. \quad (17)$$

Their real parts are nonnegative because $\cos(\alpha\pi/2) \geq 0$. For $\alpha < 1$, these elements introduce loss as well as memory; reducing the order does more than retuning the ordinary resistor. Constant phase behavior is an idealization, commonly approximated over a finite band [3]. The present results concern that mathematical model, with no claim of a fabricated fractional element.

IV. ANALYTICAL EXPECTATIONS

A. Frequency response

Let $\Omega = \omega\tau = 2\pi f\tau$ and $\theta = \alpha\pi/2$. Then

$$H_\alpha(j\omega) = \frac{1}{1 + \rho\Omega^\alpha e^{j\theta} + \Omega^{2\alpha} e^{2j\theta}}. \quad (18)$$

All orders have unity DC gain. At high frequency, the magnitude decreases as $\Omega^{-2\alpha}$, giving slopes of $-32, -36,$

and -40 dB/decade for the selected orders. The corresponding phase limits are -144° , -162° , and -180° . Thus a lower order reduces high frequency rejection as well as phase lag; it is not automatically a better low pass filter.

For $\rho = 1$, put $q = \Omega^\alpha$ and $c = \cos \theta$. Expanding the squared denominator gives

$$|H_\alpha|^{-2} = q^4 + 2cq^3 + (4c^2 - 1)q^2 + 2cq + 1. \quad (19)$$

Positive stationary points satisfy

$$2q^3 + 3cq^2 + (4c^2 - 1)q + c = 0. \quad (20)$$

This cubic distinguishes a local hump from a peak above DC gain. For $\alpha = 0.8$, the response is monotone. For $\alpha = 0.9$, a small local maximum occurs at 57.98 Hz, but its magnitude is 0.9870 (-0.1133 dB), below unity. The classical circuit has a peak of 1.1547 (1.2494 dB) at 112.54 Hz. A voltage step can overshoot even when no frequency gain exceeds the DC gain, as the $\alpha = 0.9$ case will demonstrate.

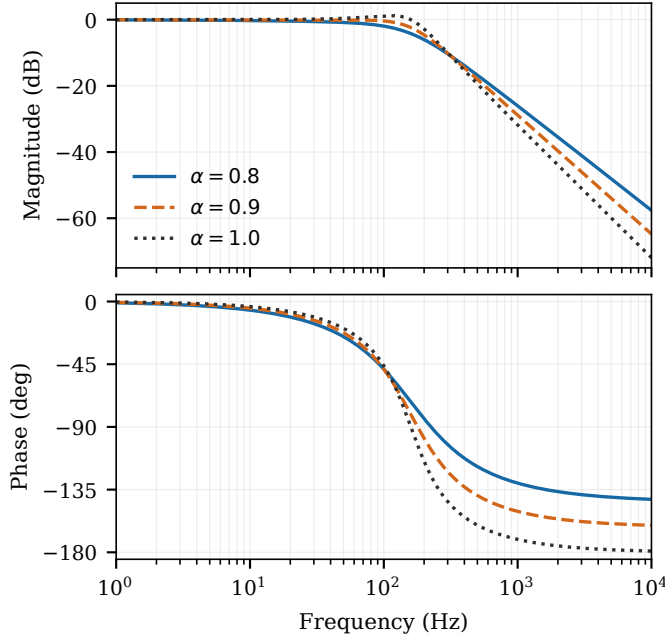


Fig. 2. Analytical frequency responses from (18), independent of the time integration grid. Frequency is in hertz; the reference scale is $\tau = 1$ ms.

B. Memory tail and settling tolerance

In normalized Laplace variable $p = s\tau$, the voltage step transform is $X(p) = H(p)/p$. Expansion at small p gives

$$\frac{1 - H(p)}{p} = \rho p^{\alpha-1} + (1 - \rho^2)p^{2\alpha-1} + \dots \quad (21)$$

Using the Laplace transform pair [2], the leading late time deficit is

$$1 - x(u) \sim \frac{\rho}{\Gamma(1 - \alpha)} u^{-\alpha}, \quad 0 < \alpha < 1. \quad (22)$$

The classical response instead decays exponentially. For the chosen $\rho = 1$, the next $u^{-2\alpha}$ coefficient vanishes. Equation (22) is an asymptotic approximation, not a bound valid at every

time. It suggests that a stringent tolerance can expose a slow residual even if the initial oscillation is small.

An approximate late time crossing of tolerance ε is

$$t_\varepsilon \approx \tau \left[\frac{\rho}{\varepsilon \Gamma(1 - \alpha)} \right]^{1/\alpha}. \quad (23)$$

This formula is useful for interpretation but is not used in place of computing settling from the response. Early band exits caused by oscillation must also be included. Comparing 2% and 1% therefore gives a more informative assessment than reporting one settling time.

V. NUMERICAL METHOD AND VERIFICATION

A. Full history backward Euler convolution

The computation discretizes the state system rather than replacing it by an ordinary polynomial transfer function. A backward Euler convolution quadrature [4] uses normalized step size h and weights from $(1 - \xi)^\alpha = \sum_{j=0}^{\infty} w_j \xi^j$:

$$w_0 = 1, \quad w_j = \left(1 - \frac{\alpha + 1}{j}\right) w_{j-1}. \quad (24)$$

For zero initial states, the Caputo and Riemann–Liouville derivative expressions agree. At $u_n = nh$, the update is

$$(I - h^\alpha A) \mathbf{z}_n = h^\alpha b r_n - \sum_{j=1}^n w_j \mathbf{z}_{n-j}. \quad (25)$$

The entire history is retained. The constant 2×2 matrix is inverted once for each order. At $\alpha = 1$, $w_1 = -1$ and later weights vanish, recovering ordinary backward Euler. Omitting the history terms for fractional orders would change the model.

The main grid is $h = 0.0025$, corresponding to $\Delta t = 2.5 \mu\text{s}$. The horizon is $u = 60$, or 60 ms, with $N = 24000$ updates per order. A direct implementation uses $O(N^2)$ work and $O(N)$ stored states. This is manageable for a course example, but fast convolution would be preferable for long records. The companion MATLAB file uses base language operations and requires no control or fractional calculus toolbox.

B. Independent checks

Three checks prevent qualitative differences from being numerical artifacts. First, the $\alpha = 1$ solution is compared with (7) at every grid point. Second, the fractional solutions are recomputed with $h = 0.01$, 0.005 , and 0.0025 , and differences are evaluated at common time samples. Third, a separate arbitrary precision inverse Laplace computation checks $X(p) = 1/[p(p^{2\alpha} + p^\alpha + 1)]$ at $u = 0.25, 1, 2, 4, 10, 20, 30, 60$. The de Hoog inversion is evaluated with 35 decimal digits using Python's `mpmath`; it does not use the convolution weights or the state update.

The reported study was executed in Python using the same recurrence supplied in MATLAB. MATLAB itself was unavailable in the execution environment, so native MATLAB runtime verification is not claimed. This distinction concerns software execution, not the definition of the model: the reference inversions, analytical frequency curve, and classical solution are independent of the MATLAB file.

TABLE I
MAXIMUM NORMALIZED VOLTAGE ERRORS

h	$\alpha = 0.8$	$\alpha = 0.9$	$\alpha = 1$
0.01	1.821e-03	2.346e-03	3.887e-03
0.005	9.126e-04	1.176e-03	1.950e-03
0.0025	4.568e-04	5.888e-04	9.768e-04

Fractional columns: eight inverse Laplace samples.
Integer column: exact solution at every grid point.

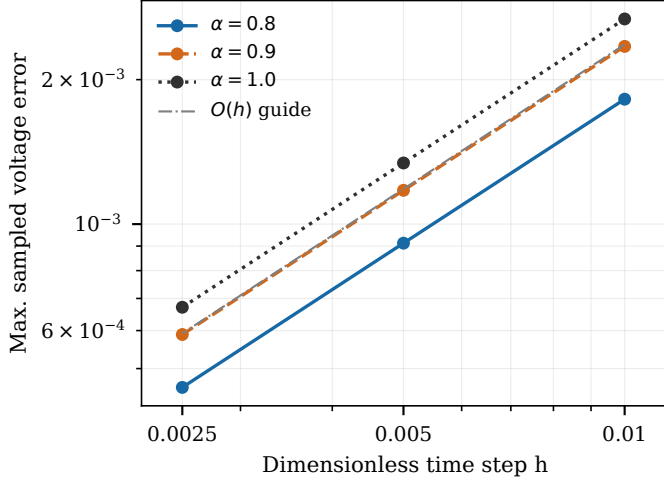


Fig. 3. Voltage errors against independent reference values as the normalized grid is refined. Integer order errors use the exact solution on the full grid; fractional errors use eight inverse Laplace samples.

Table I and Fig. 3 show decreasing errors with grid refinement. The exact peak voltage and time also agree closely at the finest step. The scheme is first order at $\alpha = 1$; fractional solutions have a weak startup singularity, so a universal convergence rate is not inferred from only three steps. Error tolerances here support the reported comparisons, rather than a claim of a new numerical method.

At the finest step, the classical all-grid voltage error is 9.768×10^{-4} V, and the overshoot error is 0.0738 percentage points. The fractional sampled discrepancies are below 6×10^{-4} V. Successive full-grid differences approximately halve when h halves. These errors support the observed comparison between orders; the final decimal places remain grid dependent.

C. Response measures

The overshoot is $M_p = 100 \max(0, \max_n x_n - 1)$. Peak time is reported only when an above-unity peak exists. Rise time is the difference between the first 90% and 10% crossings, with linear interpolation between samples. Settling time at tolerance ε is the last entry into $|x - 1| \leq \varepsilon$ after all earlier exits. The calculation searches the complete simulated record and interpolates the final band crossing.

Reported settling times are finite horizon estimates over $0 \leq t \leq 60$ ms. They are supported by stability, the tail analysis, and independent response samples, but a finite record alone is not a mathematical proof that a later band exit is

impossible. The distinction is particularly relevant to fractional models, whose memory persists beyond the plotted interval.

VI. RESULTS AND DISCUSSION

A. Voltage and current transients

Fig. 4 compares the initial voltage and current responses, and Table II gives the corresponding measures. The current scale is $V_0/Z_0 = 0.1$ A, so the plotted milliamperes follow directly from the normalized state. Positive current charges the generalized capacitor; the classical reversal is associated with energy exchange and voltage overshoot.

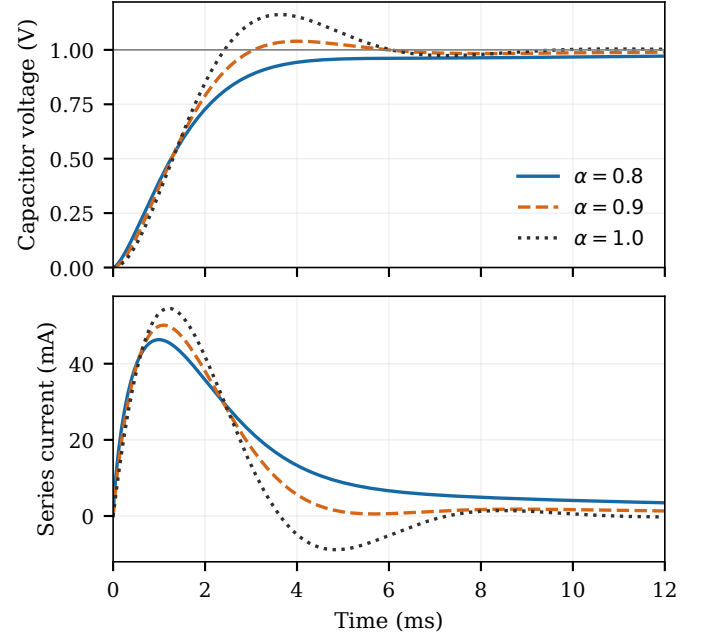


Fig. 4. Computed capacitor voltage and series current for a 1 V step. All curves use the same reference parameters and $\Delta t = 2.5 \mu s$.

TABLE II
COMPUTED STEP RESPONSE MEASURES ($h = 0.0025$)

α	M_p (%)	t_r (ms)	t_p (ms)	$t_{2\%}$ (ms)	$t_{1\%}$ (ms)
0.8	0.00	2.823	—	19.370	46.833
0.9	4.00	2.002	4.020	5.114	12.699
1.0	16.23	1.640	3.630	8.071	8.784

Last-exit settling estimates over the 60 ms record.
A dash denotes no above-unity peak in that record.

The classical circuit displays the largest overshoot. At $\alpha = 0.9$, the overshoot is about 4%, while the response at $\alpha = 0.8$ approaches the final voltage without an above-unity peak during the simulated record. This behavior is consistent with the additional loss and distributed memory in the generalized elements. It must not be interpreted as evidence that the same ordinary components spontaneously change their derivative order.

B. Two tolerances reveal a tradeoff

The $\alpha = 0.9$ circuit enters the 2% band earlier than the classical reference, because its oscillation is much smaller. At the stricter 1% tolerance, however, the classical circuit settles earlier. The fractional late time deficit persists after the initial transient appears well damped. At $\alpha = 0.8$, both settling estimates are slower, despite its absence of overshoot.

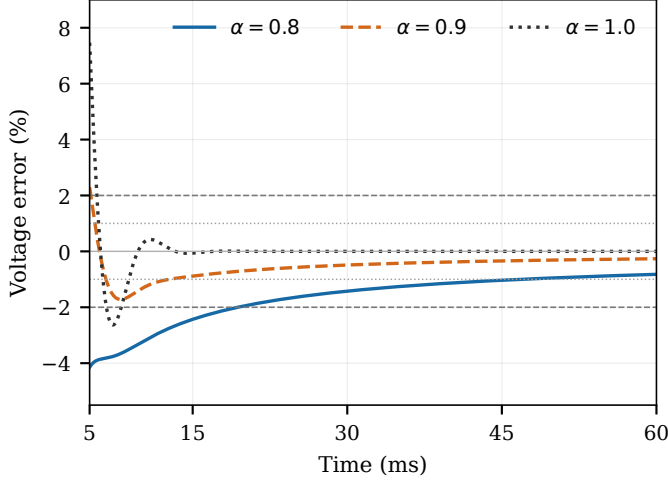


Fig. 5. Late time voltage error. Horizontal lines mark 2% and 1% tolerances. The fractional residual is visible long after the initial oscillation.

This reversal is the main engineering observation of the study. A design requirement concerned with a 2% band would favor a different response than a requirement demanding 1% accuracy. Fractional order cannot be ranked using overshoot alone. The explicit numerical horizon and tolerance are therefore part of the result, rather than incidental plotting choices.

C. Consistency with frequency behavior

The frequency curves reinforce, but do not duplicate, the transient findings. The classical 1.249 dB resonance is removed as an above-DC peak for the two lower orders. At the same time, high frequency attenuation becomes less steep. A designer seeking reduced resonance pays for that change with weaker asymptotic rejection in this fixed-parameter model. The $\alpha = 0.9$ step still overshoots even though its frequency magnitude never exceeds unity; step overshoot and frequency resonance are distinct properties.

D. Residual voltage and current

TABLE III
LATE TIME STATES ON THE MAIN NUMERICAL GRID

α	$v_C(30 \text{ ms})$ (V)	$v_C(60 \text{ ms})$ (V)	$i(60 \text{ ms})$ (mA)
0.8	0.985780	0.991787	0.861936
0.9	0.995123	0.997368	0.274770
1.0	1.000000	1.000000	$< 10^{-8}$

Table III shows why the late time interval matters. At 30 ms, the $\alpha = 0.8$ voltage is still about 1.42% below the target, while

the classical residual is already negligible on this scale. At 60 ms, the leading asymptotic deficits predicted by (22) are 0.008234 and 0.002638 for $\alpha = 0.8$ and 0.9; the computed deficits are 0.008213 and 0.002632. Their close agreement supports the memory-tail interpretation without turning the asymptotic formula into a strict error bound.

The residual current also decays more slowly in the fractional cases. It must be interpreted through the generalized capacitor law in (10), rather than by estimating $C dv_C/dt$ from the voltage graph. For $\alpha < 1$, current depends on the weighted voltage history. A nearly flat voltage curve can therefore coexist with a visible current residual. This provides a concrete signals and systems example of why a model with internal memory cannot be assessed from instantaneous slopes alone.

VII. SCOPE AND PRACTICAL LIMITATIONS

The comparison changes a constitutive order while holding L , C , R , τ , and the source amplitude fixed. It isolates the effect of the chosen model, but does not optimize resistance or fractional coefficients. Retuning parameters could alter all the rankings. Equal element orders, zero initial states, memory initialized at $t = 0$, an ideal voltage step, and linear time invariant behavior are additional restrictions.

A realistic hardware study would first measure impedance, identify a valid constant phase band, and fit L_α , C_α , and their orders to those measurements. It would then account for source resistance, coil parasitics, capacitor losses, and finite input rise time. No experimental dataset is used here. The plots and tables are computed evidence for a specified model, suitable for demonstrating signals and systems concepts and for planning a subsequent experiment.

VIII. CONCLUSION

The ordinary Laplace transform gives a clear transfer function for a dimensionally consistent fractional RLC state model. Independent numerical checks confirm its classical limit and support the computed fractional responses. For the selected circuit, reducing the order suppresses voltage overshoot and above-DC resonance, while weakening high frequency attenuation. The $\alpha = 0.9$ response improves the 2% settling estimate but worsens the 1% estimate relative to the classical circuit. This tolerance dependent tradeoff is explained by the fractional memory tail. The study provides a reproducible undergraduate analysis; experimental identification is needed before transferring its quantitative conclusions to hardware.

APPENDIX A COMPACT MATLAB IMPLEMENTATION

The following self-contained script implements (25) and plots voltage and current. Save it as `rlc_demo.m` and run it in MATLAB R2016b or later, which supports local functions in scripts. The companion `rlc_fractional_study.m` adds frequency plots, tolerance measures, grid checks, and exported data.

```
clear; clc;
aSet = [0.8 0.9 1.0];
tau = 1e-3; V0 = 1; Z0 = 10;
h = 0.0025; U = 60;
A = [0 1; -1 -1]; b = [0;1];
N = round(U/h); u = (0:N)*h;
X = zeros(3,N+1); Y = X;
for k = 1:3
    a = aSet(k);
    w = zeros(1,N+1); w(1) = 1;
    for j = 1:N
        w(j+1) = (1-(a+1)/j)*w(j);
    end
    z = zeros(2,N+1);
    G = (eye(2)-h^a*A)\eye(2);
    for n = 1:N
        hist = z(:,n:-1:1)*w(2:n+1).';
        z(:,n+1) = G*(h^a*b-hist);
    end
    X(k,:) = z(1,:); Y(k,:) = z(2,:);
end
tms = 1e3*tau*u;
subplot(2,1,1);
plot(tms,V0*X,'LineWidth',1.2);
xlim([0 12]); grid on;
ylabel('Capacitor voltage (V)');
legend('a=0.8','a=0.9','a=1.0');
subplot(2,1,2);
plot(tms,1e3*V0/Z0*Y,'LineWidth',1.2);
xlim([0 12]); grid on;
xlabel('Time (ms)'); ylabel('Current (mA)');
f = logspace(0,4,1200);
figure; hold on;
for a = aSet
    p = (1i*2*pi*f*tau).^a;
    H = 1./(p.^2+p+1);
    semilogx(f,20*log10(abs(H)));
end
set(gca,'XScale','log'); grid on;
xlabel('Frequency (Hz)'); ylabel('Gain (dB)');
legend('a=0.8','a=0.9','a=1.0');
```

For a nonzero initial state, the discretized history must be applied to $z - z_0$ with the appropriate Caputo initial contribution. The zero-state code above must not be reused

unchanged for a charged capacitor or an initial inductor current.

APPENDIX B INITIAL CONDITIONS AND A REPRODUCTION CHECK

The role of separate initial states can be seen directly in normalized Laplace variable p . For $x(0) = x_0$ and $y(0) = y_0$, transforming (11) gives

$$p^\alpha X - p^{\alpha-1}x_0 = Y, \quad (26)$$

$$p^\alpha Y - p^{\alpha-1}y_0 = R - X - \rho Y. \quad (27)$$

Eliminating Y then yields

$$X(p) = \frac{R(p) + p^{\alpha-1}y_0 + (p^\alpha + \rho)p^{\alpha-1}x_0}{p^{2\alpha} + \rho p^\alpha + 1}. \quad (28)$$

The numerator demonstrates why a zero-state transfer function does not by itself describe a charged circuit. It also explains why the state derivation is safer than assuming a composition rule for fractional derivatives.

For a quick check after running the MATLAB program, the voltage at $u = 4$ should approach the following independently inverted values as the grid is refined:

$$x_{0.8}(4) = 0.943191372, \quad x_{0.9}(4) = 1.040258345, \quad (29)$$

and $x_1(4) = 1.153122768$. Finite-step results should be compared with a tolerance of about 10^{-3} , not equality to all printed digits. The complete function also reports the classical error at every grid node and stores the states used to calculate each measure. Its output CSV files make the plotted comparisons traceable without requiring manual readings from a graph.

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